HELIUM BOIL OFF

TABLE OF CONTENTS

SECTION 2 – HELIUM BOIL OFF

2–1 INTRODUCTION

Helium boil off is the main indication of cryostat and shield cooler performance and an early warning of any problem developing within those systems. For that reason the cryogen log, shown in Table 2–1 and contained in the magnet service manual, should be maintained at regular intervals (~ 2 weeks) to monitor those systems.

The cryogen log will allow the average boil off rate to be calculated over each interval monitored. It will be helpful to graph the average boil off rates over a time as a quick reference, so any changes can be easily detected. See Graph 2–1. Ensure cryogen monitor calibration is maintained. Uncalibrated cryogen monitors will impact boil off calculations.

The rate of helium boil off is a function of thermal efficiency of the cryostat. Heat energy entering into the cryostat is intercepted by the two thermal shields which are cooled by the shield cooler. Heat transfer into the helium vessel, resulting from the difference between the shield temperatures and the liquid helium temperature (4.2 K) in the helium vessel, is removed by the evaporation and exhaust of helium.

Although boil off rates will temporarily increase during ramping, changes in barometric pressure, active shimming, certain replacement / maintenance functions and when the shield cooler is turned off, the stabilized boil off rate for properly operating cryostats / shield coolers should be well within the specification for the unit.

Boil off temperature specification for your system is shown in the Magnet Commissioning / Operating Specification Chart in the front of the Functional Check Section of the Magnet Service Manual.

If out of spec. boil off is encountered or a significant increase occurs during stabilized operation, use this section to isolate the fault.

Some service events and conditions that will increase boil off are shown in Table 2–2 (Cryogen Boil Off).

2–1 INTRODUCTION (continued)

TABLE 2–1
CRYOGEN LOG

DATE	TIME (t)	METER %	AMT (L)	CHANGE FROM PREVIOUS READING		BOIL– OFF RATE ($\Delta l \div \Delta t$)	DATE	TIME (t)	METER %	AMT (L)	CHANGE FROM PREVIOUS READING		BOIL– OFF RATE ($\Delta l \div \Delta t$)
				Δt	Δl						Δt	Δl	

2–1 INTRODUCTION (continued)

TABLE 2–2
CRYOGEN BOIL OFF

STANDARD CRYOGEN BOIL OFF AND SERVICE EVENTS THAT WILL RESULT IN EXCESS USE OF CRYOGENS

EVENT	AVERAGE OCCURANCE (PER YEAR)	CRYOGEN LOSS PER EVENT (LITERS)	AVERAGE CRYOGEN LOSS PER YEAR (LITERS)	CRYOGEN LITERS RE-FILL TO REFILL (AT 85% TRANSFILL EFFICIENCY)
SI MAGNET				
NORMAL BOIL OFF RATE		0.5 / Hour	4380	5150
P3 SOFT VACUUM	.33	1200	400	470
P3 O–RING BLOWOUT	.33	1500	500	590
FLOW IMBALANCE	1	500	500	590
* TAO	.2	750	150	175
SOFT TRANSFER LINE	.2	1200	240	285
LOW NITROGEN	.33	300	100	120
NOT USING HIGH EFFICIENCY TRANSFER LINE	1	1500	1500	1770
SII, SIII MAGNET				
NORMAL BOIL OFF RATE		0.2 / Hour	1750	2060
SHIELD COOLER REPLACEMENT	1.1	640	700	825
* TAO	Minimal			
FLOW ADJUSTMENT	Minimal			
SOFT TRANSFER LINE	.2	1050	210	250
NOT USING HIGH EFFICIENCY TRANSFER LINE	1	300	300	350

* TAO IS *THERMAL ACOUSTICAL OSCILLATION* TURBULENCE CAUSED BY DIFFERENCE IN TEMPERATURE BETWEEN LIQUID HELIUM AND ROOM TEMPERATURE.

2–1 INTRODUCTION (continued)

TABLE 2–2 (continued)
CRYOGEN BOIL OFF

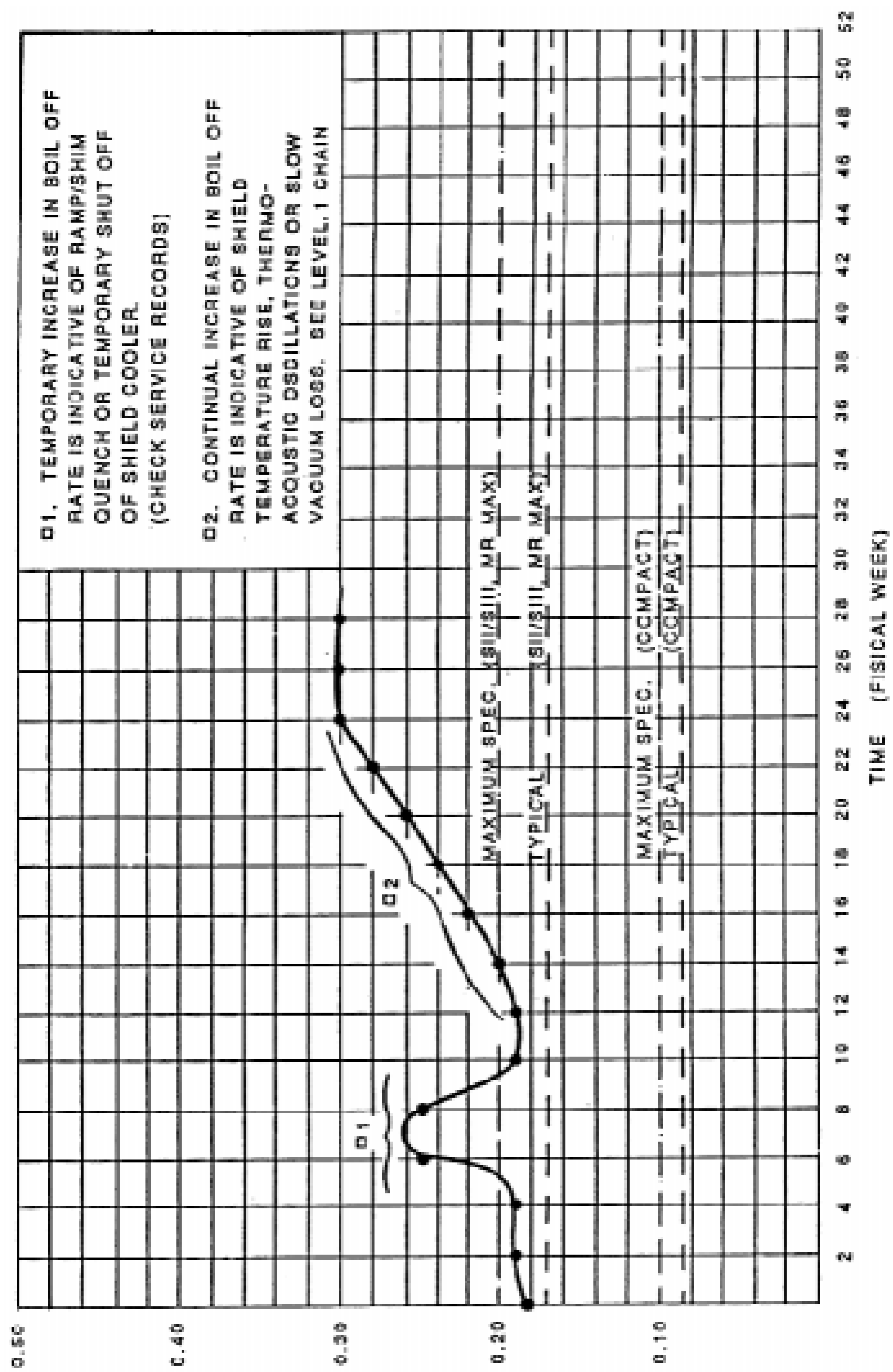
STANDARD CRYOGEN BOIL OFF AND SERVICE EVENTS THAT WILL RESULT IN EXCESS USE OF CRYOGENS

EVENT	AVERAGE OCCURANCE (PER YEAR)	CRYOGEN LOSS PER EVENT (LITERS)	AVERAGE CRYOGEN LOSS PER YEAR (LITERS)	CRYOGEN LITERS RE-FILL TO REFILL (AT 85% TRANSFILL EFFICIENCY)
MAX MAGNET				
NORMAL BOIL OFF RATE		0.2 / Hour	1750	2060
SHIELD COOLER REPLACEMENT	1.1	210	235	275
* TAO	Minimal			
FLOW ADJUSTMENT	Minimal			
SOFT TRANSFER LINE	.2	1050	210	250
NOT USING HIGH EFFICIENCY TRANSFER LINE	1	300	300	350
MAX MOBILE MAGNET				
NORMAL BOIL OFF RATE		.37 / Hour	3240	3815
SHIELD COOLER REPLACEMENT	1.1	210	235	275
* TAO	Minimal			
FLOW ADJUSTMENT	Minimal			
SOFT TRANSFER LINE	.2	1050	210	250
NOT USING HIGH EFFICIENCY TRANSFER LINE	1	300	300	350

* TAO IS *THERMAL ACOUSTICAL OSCILLATION* TURBULENCE CAUSED BY DIFFERENCE IN TEMPERATURE BETWEEN LIQUID HELIUM AND ROOM TEMPERATURE.

2–1 INTRODUCTION (continued)

GRAPH 2–1
BOIL OFF RATE VS. TIME

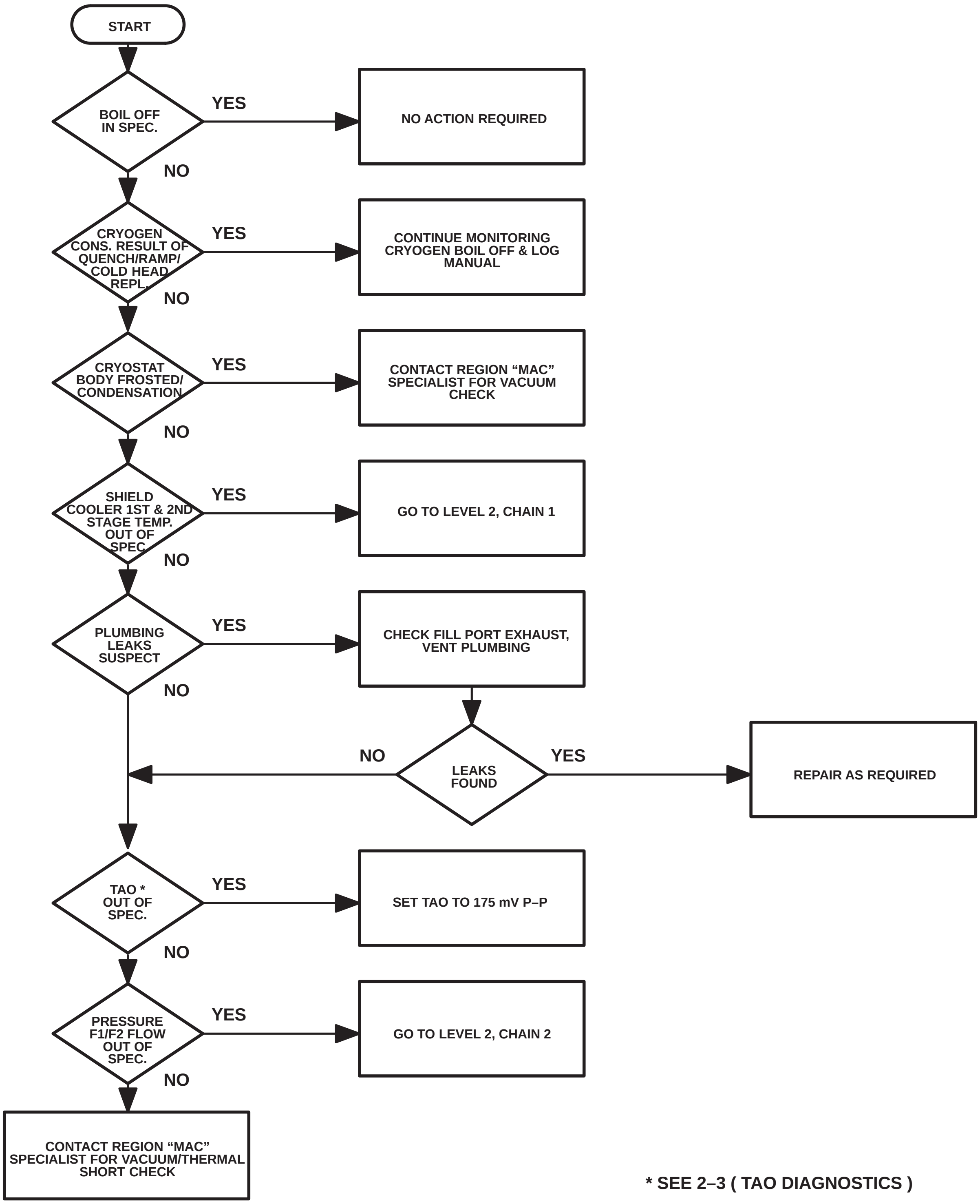


BOIL OFF RATE (LITERS / HOUR)

2–2 HELIUM BOIL OFF FAULT ISOLATION

LEVEL 1 FAULT ISOLATION CHAIN – HELIUM BOIL OFF

Specifications referred to in the Fault Isolation Chain are given in the Magnet Commissioning / Operating Specification Chart in the front of the Functional Check Section of the Magnet Service Manual.

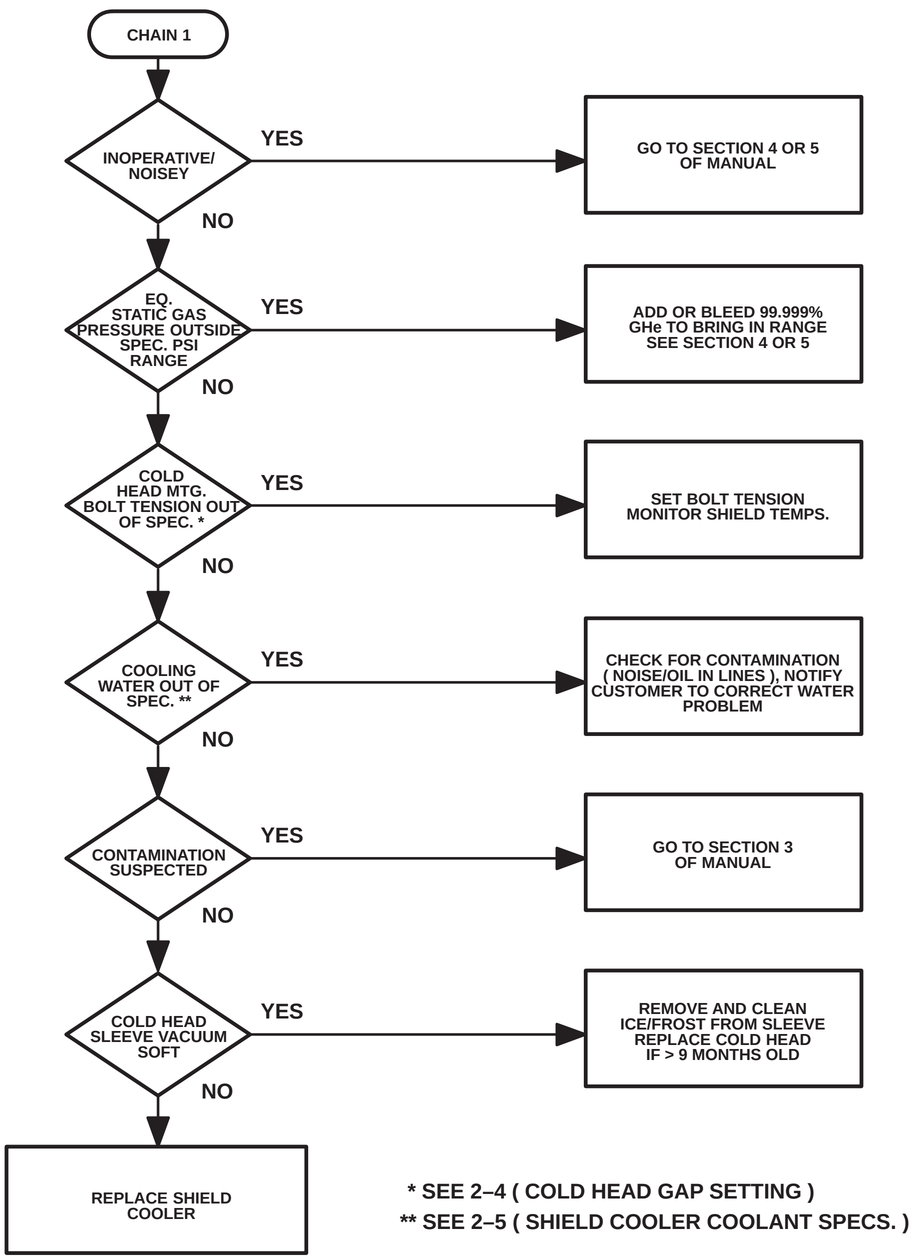


LEVEL 1 – HELIUM BOIL OFF FAULT ISOLATION CHAIN
ILLUSTRATION 2–1

2–2 HELIUM BOIL OFF FAULT ISOLATION (continued)

LEVEL 2 FAULT ISOLATION CHAIN – HELIUM BOIL OFF
CHAIN 1 – SHIELD COOLER

Note
Use Level 2, Chain 1 to rapidly isolate problems with the shield cooler system, mounting or cooling water.



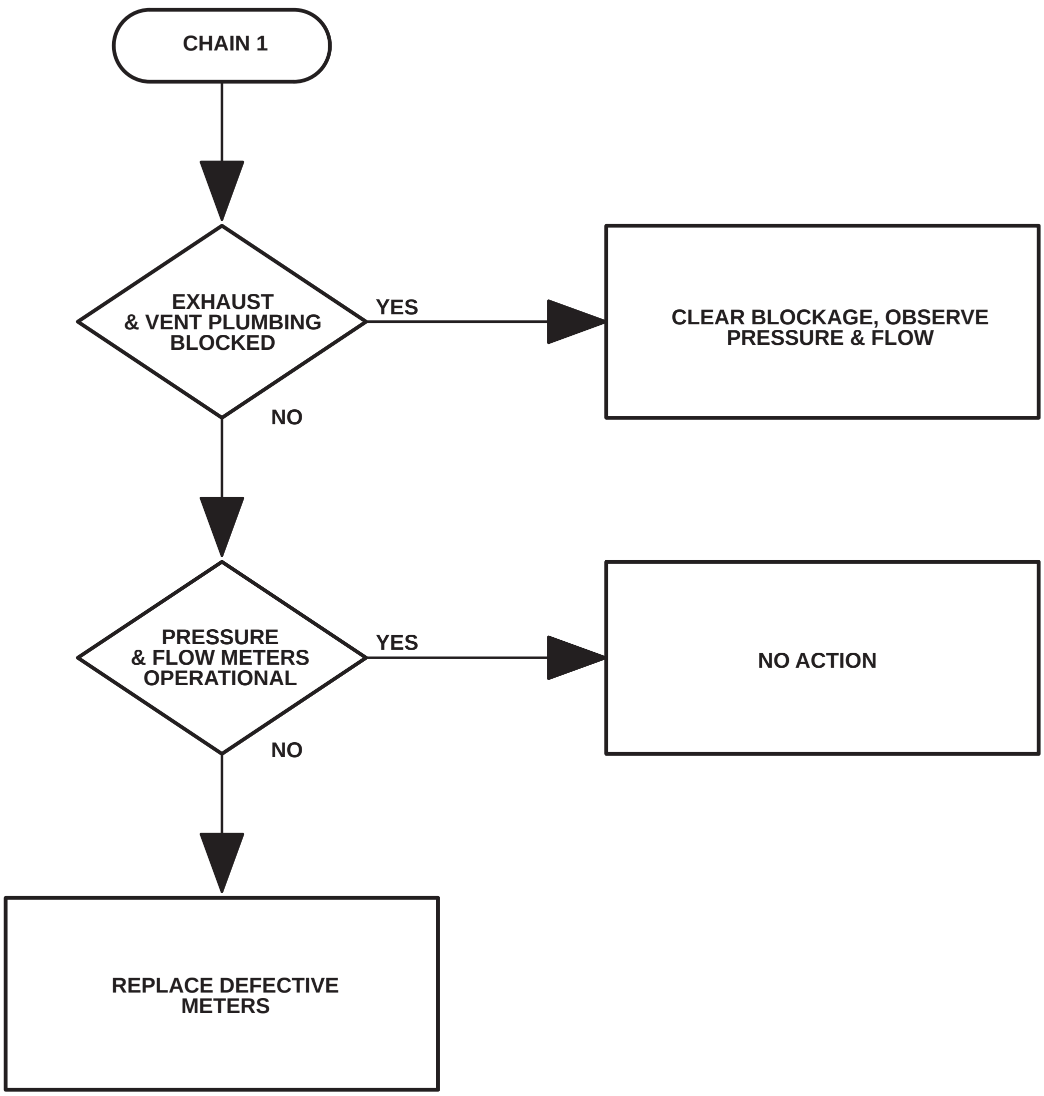
LEVEL 2 – HELIUM BOIL OFF FAULT ISOLATION CHAIN 1
ILLUSTRATION 2–2

2–2 HELIUM BOIL OFF FAULT ISOLATION (continued)

LEVEL 2 FAULT ISOLATION CHAIN – HELIUM BOIL OFF
CHAIN 2 – CRYOSTAT PRESSURE / FLOWS

Note

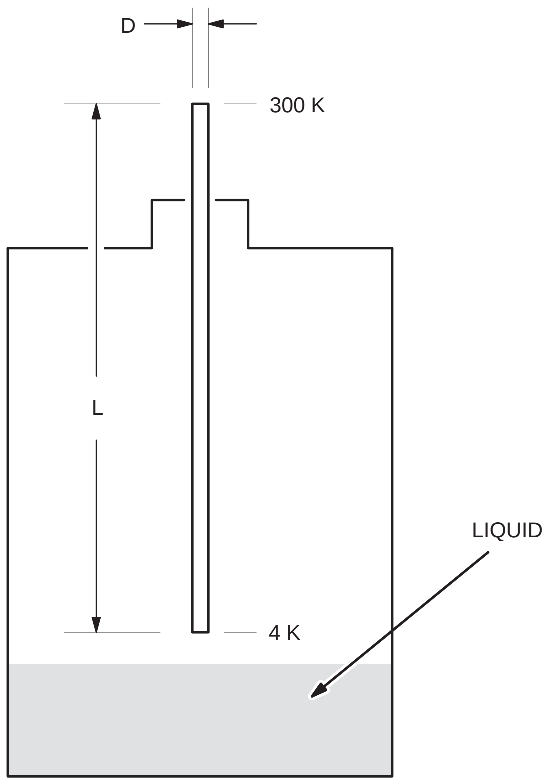
Excessive cryostat pressure / flows are generally a result of problems identified in the Level 1 Chain for helium boil off. This Chain deals with the exception conditions that may be encountered when all elements in Chain 1 check out okay.



LEVEL 2 – CRYOSTAT PRESSURE / FLOWS FAULT ISOLATION CHAIN 2
ILLUSTRATION 2–3

2–3 THERMOACOUSTIC OSCILLATIONS (TAO's) DIAGNOSTICS

2–3–1 THERMOACOUSTIC OSCILLATIONS (BACKGROUND INFORMATION)



CHARACTERIZED BY:

(1) LARGE ASPECT RATIO (L / D)
(LONG SKINNY TUBE) L / D > 50

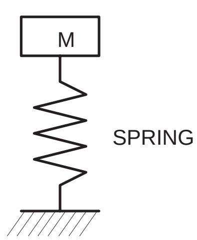
(2) TEMPERATURE GRADIENT 300 – 4 K

(3) OCCUR SPONTANEOUSLY AS SINE WAVE FREQUENCY / AMPLITUDE

RESULT:

HEAT LEAKS UNDER SOME CONDITIONS CAN INCREASE BY A FACTOR OF 1000.

SPRING MASS ANALOGY



M : HELIUM GAS

SPRING : TEMPERATURE DIFFERENCE 300 – 4 K
PHYSICAL CHARACTERISTICS OF TUBE

AS FREQUENCY ↑ HEAT LEAK ↑ 0 – 80 HZ.

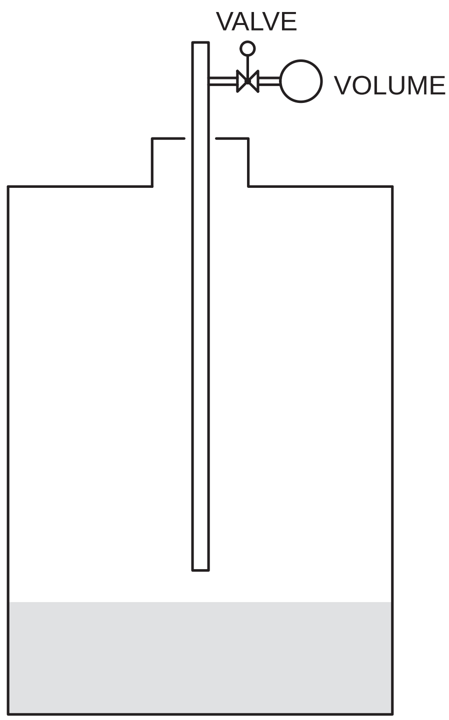
AS AMPLITUDE ↑ HEAT LEAK ↑ 0 – 6 PSIG

2–3–2 THERMOACOUSTIC OSCILLATION LOCATIONS SII, SIII & MR MAX

LOCATION	TUNING MECHANISM	COMMENTS
(1) G10 SLEEVE	SURGE BOTTLE / VALVE	STD. ON EACH MAGNET
(2) SHIM LEAD	SURGE BOTTLE / VALVE	OPTIONAL KIT
(3) FILL LINE	SURGE BOTTLE / CAPILLARY TUBE	STANDARD / NO ADJUST

2–3–3 THERMOACOUSTIC OSCILLATION ELIMINATION (BACKGROUND INFORMATION)

(BY SURGE BOTTLE / VALVE IN PARALLEL WITH TUBE)



<u>BOTTLE / VALVE:</u>

CHANGE PHYSICAL CHARACTERISTICS
OF TUBE SO TAO's ARE TUNED OUT.

TAO's MEASURED USING TRANSDUCER / OSCILLOSCOPE / DIGITAL VOLT METER:
AMP $\leq$ 250 MV [0.5 PSIG] TAO's WILL NOT SIGNIFICANTLY EFFECT BOIL OFF

NOTE: DIGITAL VOLTMETER MUST BE ON AC SCALE

2–3–4 THERMOACOUSTIC OSCILLATION ADJUSTMENT SII, SIII, MR MAX

Description:

The detailed functional checks / settings described below should be performed in conjunction with the Shield Cooler Checks (Functional Checks, Section 6) or whenever excessive Cryostat pressure and boil off are encountered. These checks will establish if the “TAO’s” in the Cryostat and Vertical Stack Gap are within specification and provide a method for optimizing the Surge Tank Valve settings. These checks / settings should be performed by a General Electric Authorized Service Representative.

A TAO Monitor Kit (46–281406G1) Fill Line Adapter (46–281232G1) and Digital Voltmeter or Oscilloscope are required for this procedure. If the magnet is at field, a Digital Voltmeter may be used in the Exam Room or an Oscilloscope may be set up outside the Exam Room and connected to the TAO Monitor.

Some units may have a “TAO” Assembly on the Shim Lead. Use the procedure and values in this section with adaptor tool (46–281232G2) for Functional Check of that assembly.

WARNING!

DO NOT BRING AN OSCILLOSCOPE INTO THE EXAM ROOM IF THE MAGNET IS AT FIELD.

Procedure:

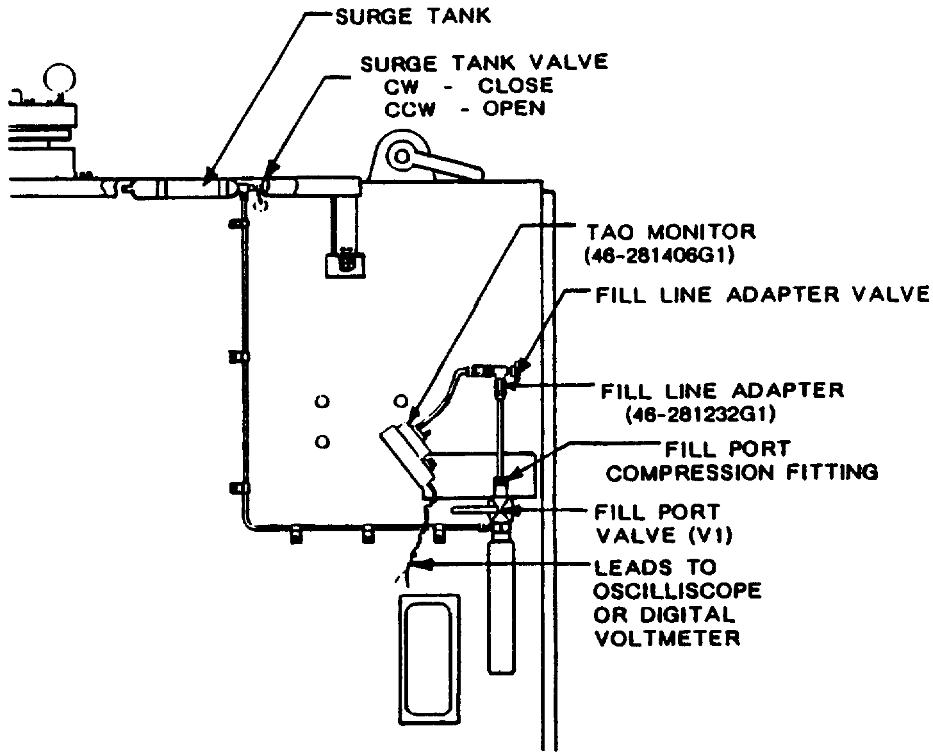
1. Observe Cryostat Pressure Gauge, if pressure is greater than 0.5 psig, slowly open Cryostat Vent Valve (V2) to bring pressure within 0.5 psig, then close (V2) and allow pressure to stabilize.
2. Loosen Compression Fitting on Cryostat Fill Port above Fill Valve (V1) and remove cap / plug.
3. Insert Fill Line Adapter Tube into Fill Port and tighten Compression Fitting. See Illustration 2–4.
4. Connect other end of Fill Line Adapter (smaller plastic tube) over top port of the TAO Monitor Pressure Transducer.

Note

An Oscilloscope or Digital Voltmeter may be used to monitor the signal from the TAO Monitor. The Oscilloscope will display the signal, providing a more positive measurement; However, it must be located outside of the Exam Room if the magnet is at field. Digital Voltmeter must be set on AC scale.

5. Connect the TAO Monitor Leads to the Oscilloscope or Digital Voltmeter. Set theOscilloscope Amplitude to 10mV / cm or the Digital Voltmeter Scale to 260 mVAC.

2–3–4 THERMOACOUSTIC OSCILLATION ADJUSTMENT SII, SIII, MR MAX (continued)



CRYOSTAT TAO ADJUSTMENT SET UP
ILLUSTRATION 2–4

6. Slowly open Fill Port Valve (V1) then open the valve on the Fill Line Adapter and read amplitude of signal. Signal amplitude is measured by the " peak to peak" reading on the Oscilloscope or the "max. – min." readings on the Digital Voltmeter.

Note

If no signal is present, replace the battery in the TAO Monitor. If signal is stlll absent, check replaced battery, circuit and TAO Monitor by putting a small positive pressure at tube connection.

7. Compare readings with the maximum allowable specification value of 175mV. If signal amplitude is greater than 175mV, adjust Surge Tank Valve (CCW than CW) to minimize the signal amplitude. See Illustration 2–4.
8. Record results for future reference (____).
9. Close V1 and the Fill Line Adapter Valve. Disconnect set up leaving leads connected to the Oscilloscope or Digital Voltmeter.
10. Cap Fill Port and tighten Compression Fitting.

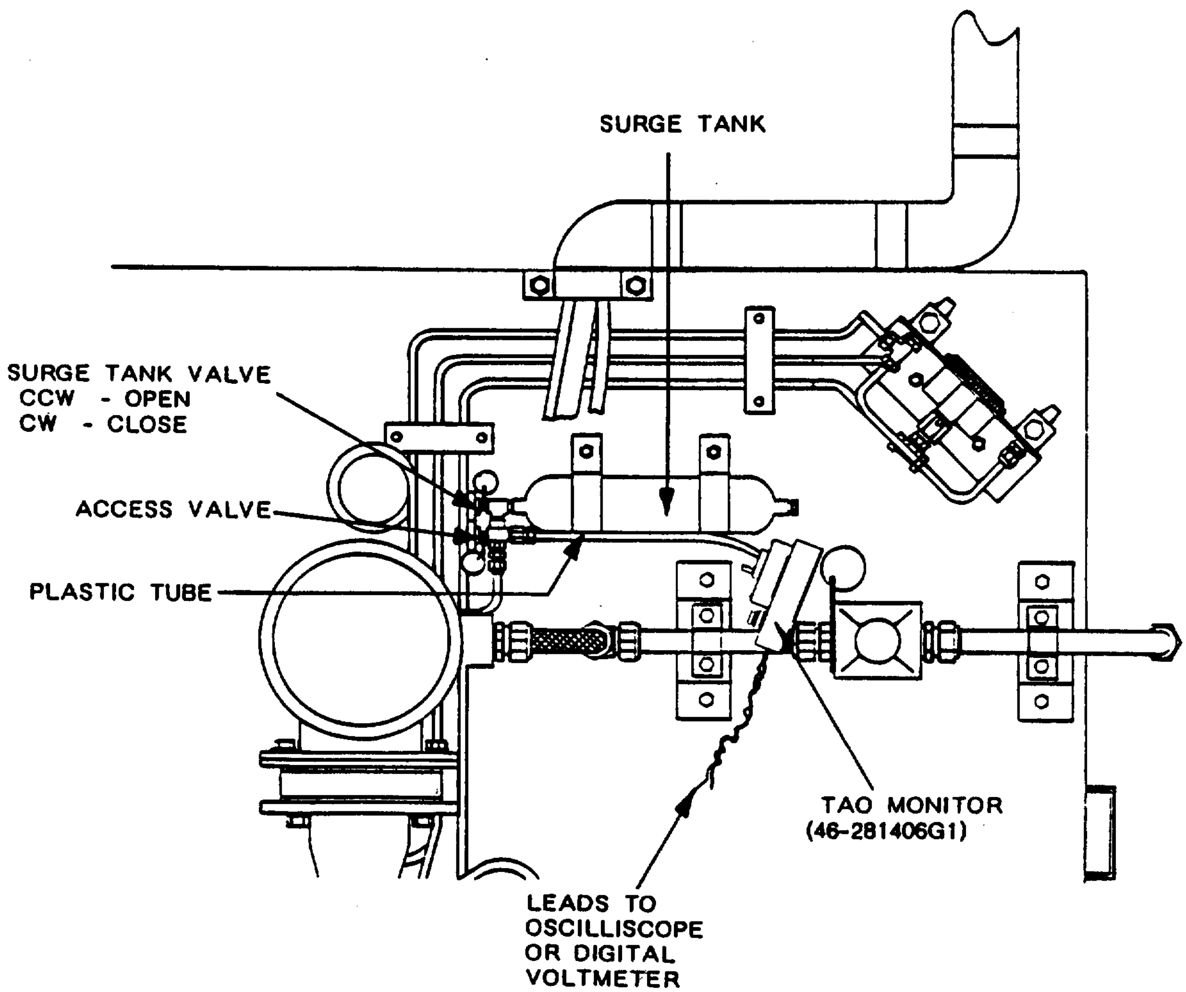
2–3–4 THERMOACOUSTIC OSCILLATION ADJUSTMENT SII, SIII, MR MAX (continued)

11. Remove TAO Monitor fron Fill Line Adapter and connect (P2) of the TAO Monitor Pressure Tracducer to the plastic tube next to the Surge Tank for the Vertical Penetrtion Gap. See Illustration 2–5.

12. Slowly open access Valve at end of plastic tube and repeat Step 7 of this procedure.

13. Record results for future reference (________________________________).

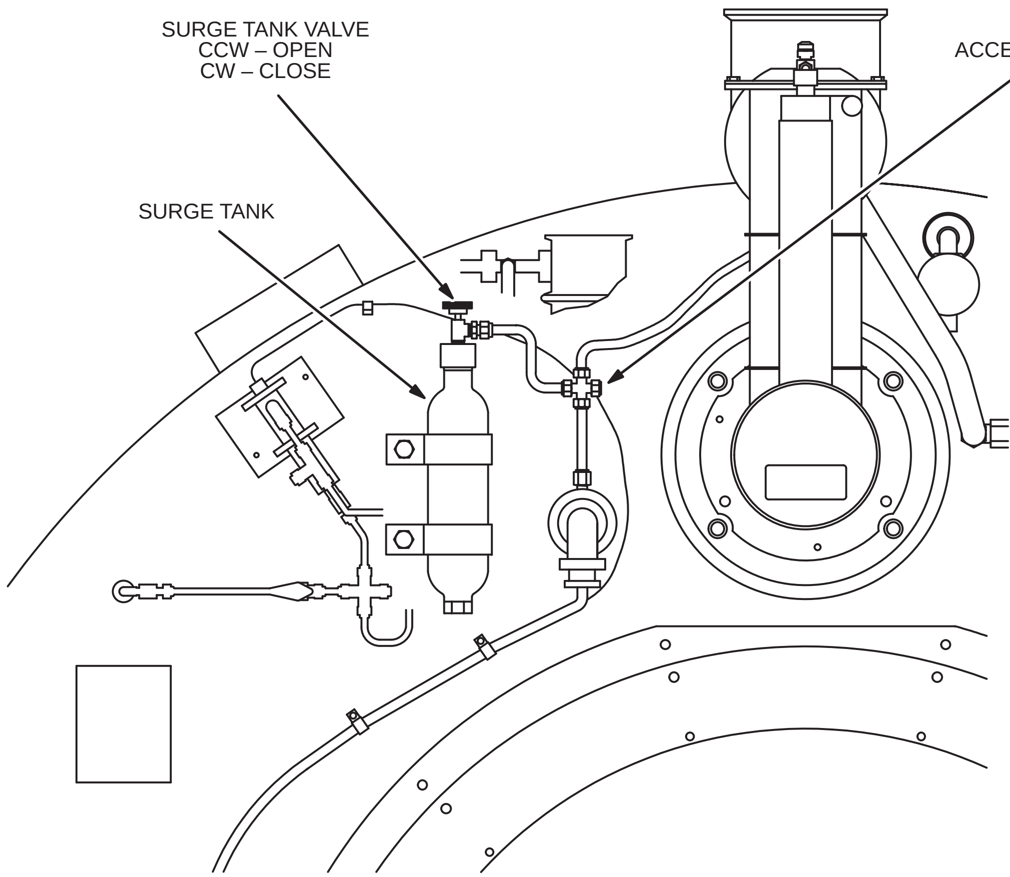
14. Close Access Valve and remove set up.



VERTICAL PENETRATION GAP TAO ADJUSTMENT SET UP
ILLUSTRATION 2–5

2–3–5 THERMOACOUSTIC OSCILLATION ADJUSTMENT SI

TAO adjustments on SI Magnets are performed in the same manner as in SII, SIII and MR Max Magnets shown in the previous Section 2–3–4. See Illustration 2–6 for configuration reference.



SEVERAL ITEMS OF PLUMBING HAVE BEEN REMOVED FOR CLARITY

SI TAO'S ON SECOND PENETRATION
ILLUSTRATION 2–6

2–4 SETTING COLD HEAD TENSION

Description:

During magnet installation or anytime the Cold Head has been shut off for a considerable length of time (days), the Cold Head will contract as it begins operating and cooling down. The tightness of the Cold Head Mounting Bolts will need to be checked, and they may need to be adjusted periodically to ensure that good contact is maintained between the Cold Head and the Cold Head Sleeve.

Procedure:

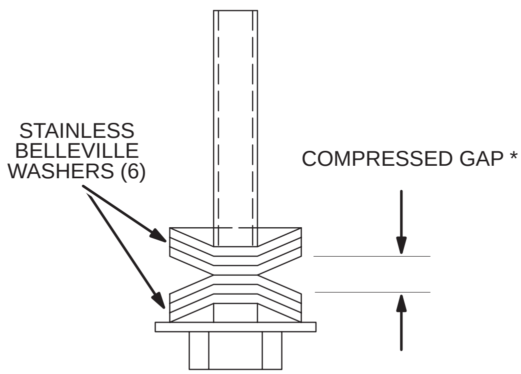
1. Monitor the Cold Head first and second stage temperatures:
   The operating temperatures should be within the following ranges:

   First Stage: 32 – 60 K
   Second Stage: 7 – 17 K

   The factory temperatures that were found for your magnet in the factory are recorded in the Acceptance Test Report (ATR) found in the Data Sheet Section of your manual.

2. If temperatures are higher than those values, check and adjust Belleville washer gap in conformance with Illustration 2–7.

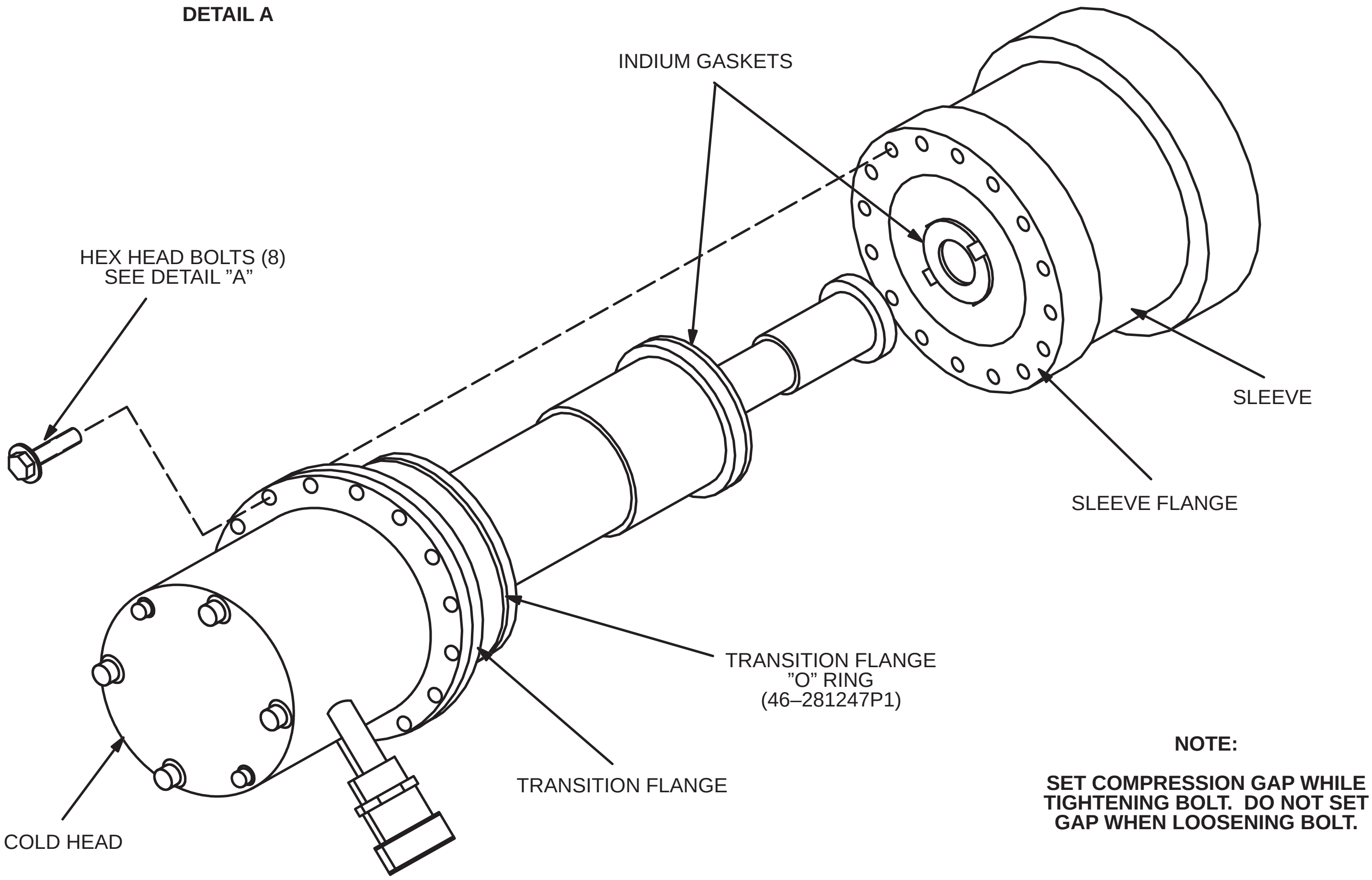
2–4 SETTING COLD HEAD TENSION (continued)



DETAIL A

* Washer Type	P/N	COMPRESS GAP SETTING
Stainless	46–281387P1	0.010” – 0.015”
Be. Cu.	46–252317P1	0.003”

* Measure gap at edge of washers.
* Be Cu washers are bronze colored.
  Stainless washers are dark gray.



COLD HEAD MOUNTING BOLT GAP SETTING
ILLUSTRATION 2–7

2–5 COOLANT SPECIFICATION FOR SHIELD COOLER COMPRESSORS

2–5–1 INTRODUCTION:

Shield cooler compressors require a circulating coolant to dissipate heat output. It is essential for compressor life, that adequate coolant is provided. Coolants which do not meet this specification will cause compressor overheat problems and result in both shut downs and a significant reduction in compressor life!

Obtain a laboratory analysis of the intended coolant prior to the magnet installation, to make sure the coolant is in compliance with this specification. “Open Systems” using “tap” water should be sampled at yearly intervals. If a Water Tee P/N 46–318696G1 is installed, the Shield Cooler Test Kit P/N 46–318784G1 can be used to test the water samples. Refer to Section 2–6 for Water Tee Installation/Operation. Refer to Section 2–7 for Shield Cooler Test Kit Operation.

2–5–2 COOLANT SOURCE & CONNECTION:

The specified coolant is water which meets the properties in Section III.

Systems which are exposed to temperatures below 40 degrees Fahrenheit require the addition of a good industrial grade antifreeze in a ratio sufficient to prevent freezing at the exposure temperature.

CAUTION

Do not use antifreeze concentrations beyond that which is absolutely necessary to prevent freezing, as heat transfer characteristics will be unnecessarily decreased.

Antifreeze which contains silicates will cause pump seal abrasion and leaks in chillers over time.

WARNING!

ANTIFREEZE IS POISONOUS TO AQUATIC, ANIMAL AND HUMAN LIFE. BECAUSE OF ITS SWEETNESS, ANIMALS MAY BE DRAWN TO IT. CONSEQUENTLY IT IS CLASSIFIED AS HAZARDOUS BOTH FOR SHIPMENT AND AS A WASTE. THEREFORE, IT MAY NOT BE SHIPPED BY UNTRAINED PEOPLE. SHIELD COOLER COMPRESSORS WHICH CONTAIN ANTIFREEZE MUST BE DRAINED BEFORE BEING RETURNED TO THE VENDOR. MAKE ARRANGEMENTS WITH THE HOSPITAL MAINTENANCE DEPARTMENT FOR DISPOSAL.

Use flexible hose of 0.5 inch (12.7 mm) inside diameter and 1.0 inch (25.4 mm) adjustable compression clamps for the inlet and outlet connections. Make sure all connections are “leak tight”.

2–5–3 COOLANT PROPERTIES:

1. ph level (6–8).

Note
Do not use Distilled Water. Distilled Water is acidic (4 – 6 ph).

2. HARDNESS (LESS THAN 100 ppm of CALCIUM / MAGNESIUM CARBONATE)

Note
Hard water will produce calcium deposits in the compressor and decrease cooling efficiency.

3. ALKALINITY (16 ppm MINIMUM)

4. SUSPENDED MATTER (LESS THAN 10 mg PER LITER)
(LESS THAN 150 MICRON PARTICLE SIZE)

Note
To meet the specification for suspended matter, install a 100 – 150 micron filter at the inlet to the compressor.

2–5–4 COOLANT CAPACITY:

EQUIPMENT	INLET TEMPERATURE RANGE ° F (° C)	INLET PRESSURE psi (KPa)	REQUIRED FLOW RATE gal/min (liter/min)	PRESSURE DROP[1] psi (KPa)	TEMPERA-TURE RISE[2] Δ° F (° C)	TYPICAL HEAT OUTPUT BTU/Hr (Watts)	MAXIMUM HEAT OUTPUT BTU/Hr (Watts)
Shield Cooler Compressor	40 – 70	Minimum 30 (207)	Minimum 2.2 (5.7)	7 (48)	15 (8.3)	13652	17748
	(5 – 20)	Maximum 125 (862)	Maximum 4.0 (15.1)	32 (221)	5.5 (3.1)	(4000)	(5200)

Note 1 Pressure drop across equipment is given for minimum and maximum recommended flow rates as indicated.
2 Water temperature rise is given for minimum and maximum recommended flow rates as indicated.

2–5–5 PLANNED MAINTENANCE:

Flush and refill coolant systems at two year intervals and whenever coolant problems are encountered.

Coolant systems should be routinely checked for leaks and that reservoirs are filled to capacity. Leaks and low coolant level can result in suspended air in the coolant and foaming, both conditions will greatly reduce cooling efficiency.

2–6 WATER TEE ASSEMBLY INSTALLATION / OPERATION INSTRUCTIONS

DESCRIPTION

The Water Tee Assembly, P/N 46–318696G1, is installed in line with the Compressor Water Supply Inlet. The Water Tee functions by diverting Compressor Cooling Water into a clean container where the “sample” can be tested for high or low PH levels using the PH tester P/N 46–318648P1, or for Hardness using the Dissolved Solids Tester P/N 46–318649P1. These two testers are included in the Shield Cooler Test Kit P/N 46–318784G1. The Water Tee must be ordered separately and is not included in this kit.

PROCEDURE

1. Turn off Compressor unit.
2. Turn off water flow to Compressor.

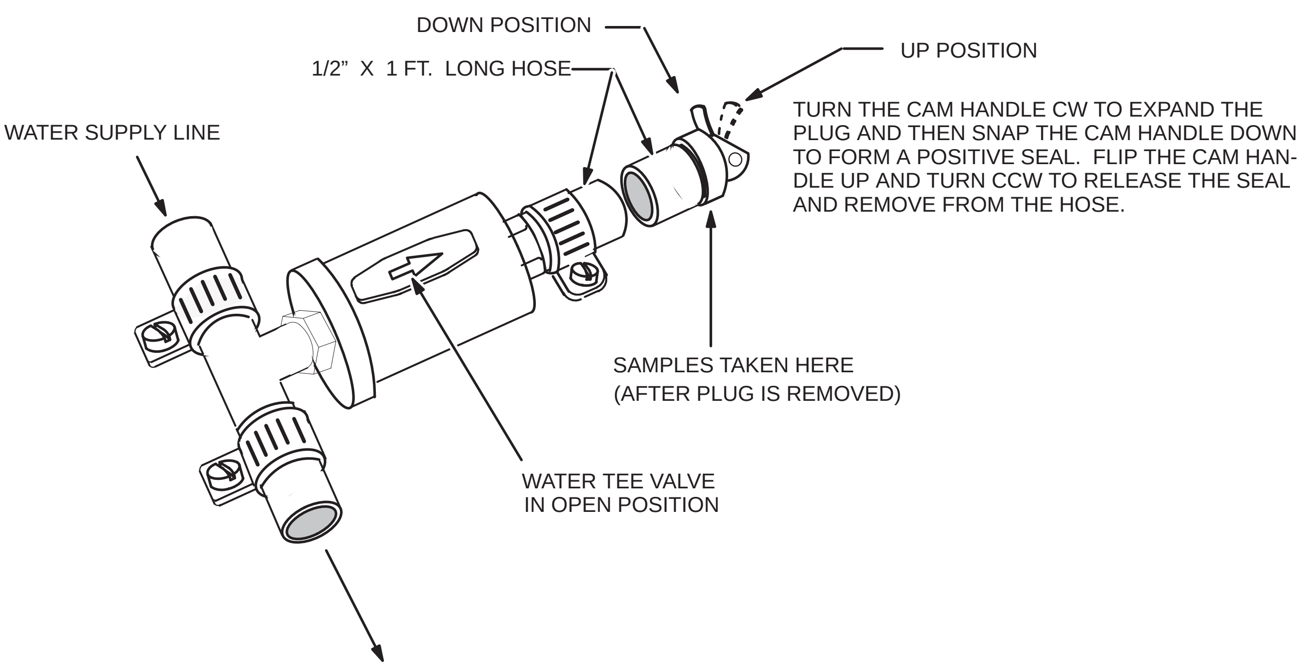
Note

Water Tee is designed to work with 1/2 inch water supply lines. Make sure water supply line is 1/2 inch (Inside diameter) before cutting the supply line.

Note

Try to allow as little water as possible to spill from the Water Supply Line after it is cut. This is to prevent air bubbles from forming in the line.

3. Cut the Water Supply Line at a convenient location for taking samples. Try to locate the Water Tee where water cannot come in contact with electronic equipment in the event a leak should occur.
4. Slide a Clamp over either cut end of water supply line. See Illustration 2–8.



WATER TEE COMPONENTS
ILLUSTRATION 2–8

2–6 WATER TEE ASSEMBLY INSTALLATION / OPERATION INSTRUCTIONS (continued)

Note

Close the Valve on the Water Tee before installing.

5. Install Water Tee into water supply line. Make sure water lines fit close to the Water Tee body as shown in Illustration 2–8.
6. Make sure the Water Tee Valve is off. See Illustration 2–8.
7. Make sure the Water Tee Quick Snap Expandable Plug is installed and locked into place.
8. Turn on Compressor Water Supply Line. Make sure no water leaks from the Water Tee.
9. Turn on compressor.

Note

Compressor does not need to be turned off to take water samples. Compressor Water Supply, also, does not need to be turned off to take a water sample.

10. To take water samples, place a small, clean container at the Water Tee Outlet, pull the Water Tee lever up and turn counterclockwise then remove the Quick Snap Expandable Plug. Container must be able to hold at least 1 1/2 inches of water. See Illustration 2–8.
11. Turn the Water Tee Valve in the direction of water flow, to the sample cup, to open valve and collect sample.
12. Turn the Water Tee Valve back to the off position to stop water flow to the sample container.
13. To replace the Quick Snap Expandable Plug, insert the plug back into the water tee hose, turn the lever clockwise to expand the plug against the sides of the tube, then snap the lever down to form a positive seal.

CAUTION

Make sure that Water Tee Valve is completely shut off after taking samples. Check for leaks before leaving the site.

2–7 SHIELD COOLER TEST KIT 46–318784G1 OPERATING INSTRUCTIONS

Description

The Shield Cooler Test Kit includes three pieces of test equipment: 1) UV Light – for locating compressor oil contamination in the cold head or helium lines. 2) pH Tester – for testing the pH level in the compressor cooling water. 3) Dissolved Solids Tester – for testing the hardness of the compressor cooling water. The Water Tee, P/N 46–318696G1, must be installed in line with the compressor supply line in order to use the pH tester or the Dissolved Solids Tester. The pH Tester and the DiST Tester are unrepairable items. The UV Light can be sent back to SPO for repair.

Equipment Included In Kit

- Dissolved Solids Tester – 46–318649P1
- pH Meter – 46–318648P1
- UV Light – 2100976
- UV Light safety goggles – 46–318829P1
- pH calibration buffer – 2100662
- Dissolved Solids Tester calibration buffer – 2100661
- Sample/buffer containers – 2100660
- Kit operating instructions – 2101994

Equipment Not Included In Kit

- Distilled water – for cleaning electrodes and sample and buffer containers.
- Water Tee – P/N 46–318696G1 – Enables easy access for taking compressor cooling water samples.
- Shield Cooler Installation Kit – P/N 46–281088G3 (needed with the UV Light to locate oil contamination).

2–7–1 pH TESTER CALIBRATION

Description

The pH meter has a range of 0.0 to 14.0 pH. The accuracy of the pH Meter is ± 0.2 pH. To change the batteries, carefully pull out the battery case and replace batteries (3 x 1.4volt Duracell MP675H or equivalent). See illustration 2–10 for location of Calibration Trim Pot and Battery Case.

2–7–1 pH TESTER CALIBRATION (continued)

Note

Containers for both the Calibration Solution and water sample are included in the Shield Cooler Test Kit.

1. Clean pH Tester Electrode and Container (included in the Shield Cooler Test Kit) with distilled water, then dry.
2. Add enough Calibration Solution (GE P/N 2100662), into the clean container, to cover the pH Tester Electrode.
3. Establish the pH value of the Calibration Solution for the pH Tester.
4. Immerse the pH Meter into the solution and adjust the Calibration Trimmer, located on the rear of the meter, to indicate the value determined in Step 3.
5. Turn the meter off using the ON/OFF switch.
6. Rinse the meter electrode and calibration solution container in distilled or deionized water, then dry.
7. Replace the protective cap.

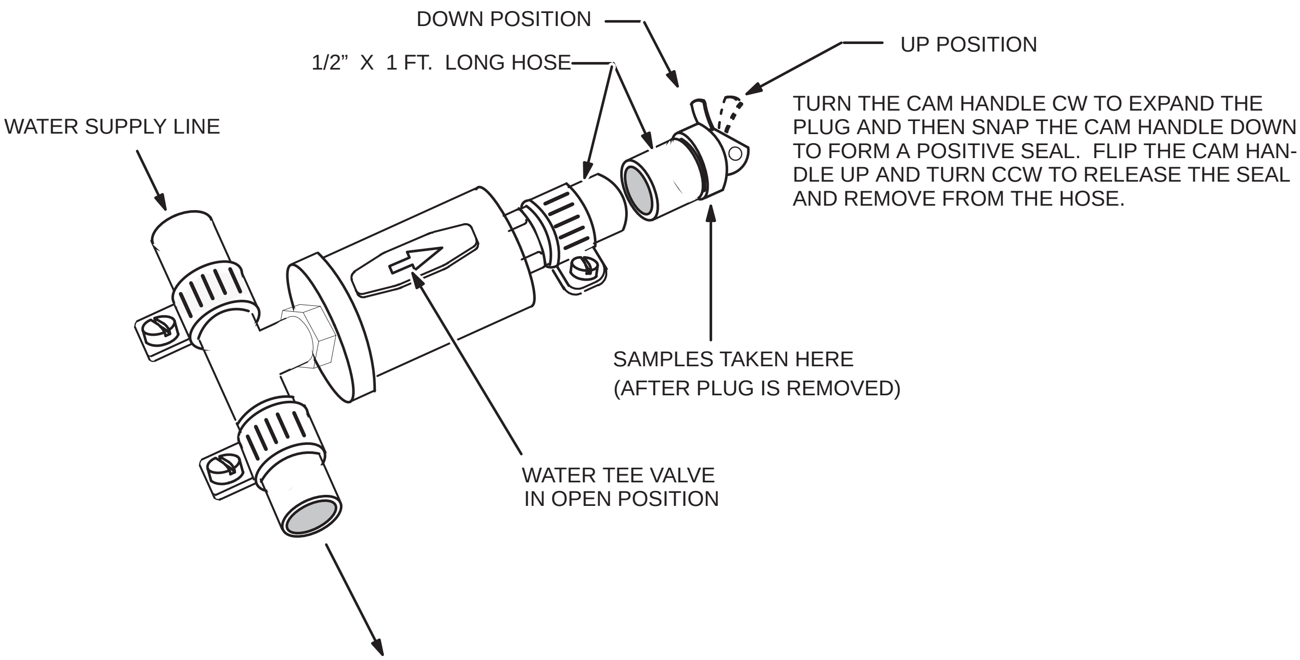
2–7–2 pH TESTER OPERATION

1. Remove the protective cap from the bottom of the pH meter.
2. Turn on the pH meter using the ON/OFF switch.
3. Place a clean Sample Container (contained in the Shield Cooler Test Kit) at the end of the Water Tee Sample Line. See Illustration 2–9.

Note

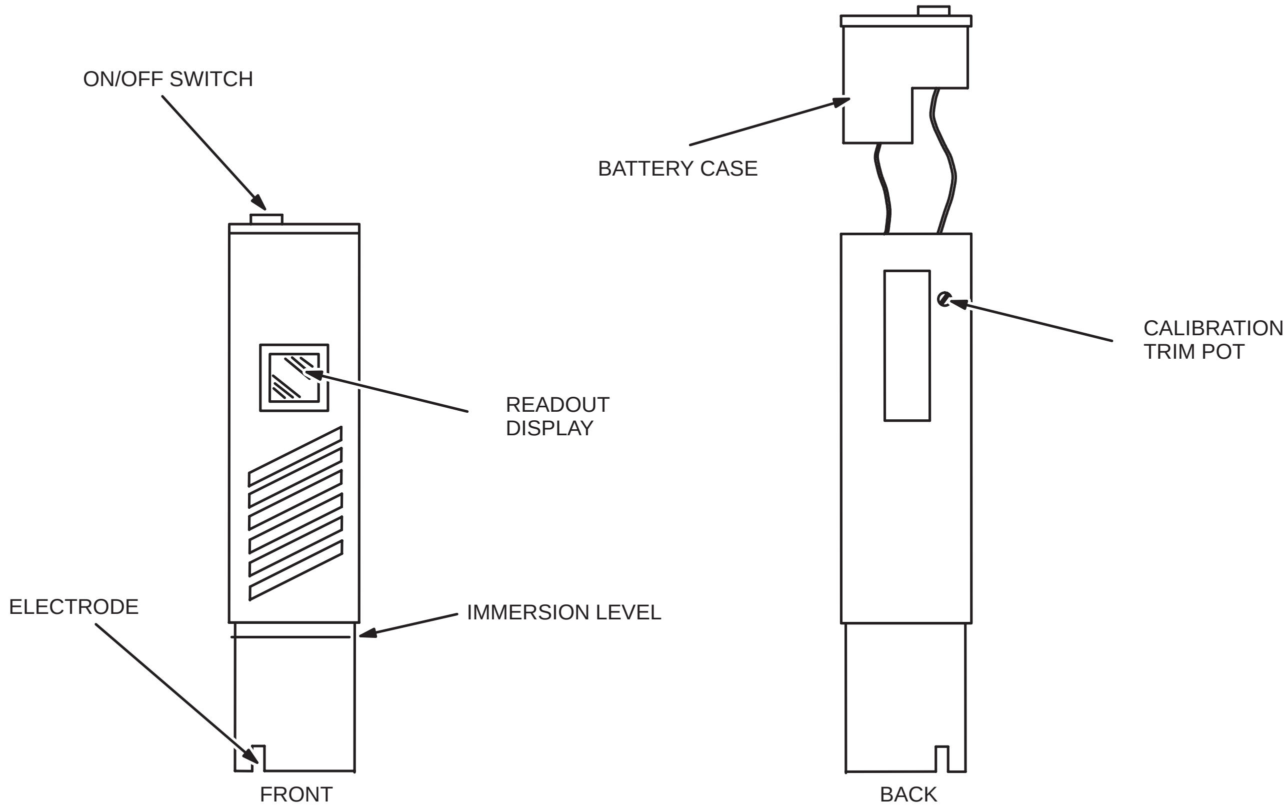
The Water Tee is designed with a safety Quick Snap Expandable Plug. To take a water sample, pull the lever up and turn counterclockwise and remove the Quick Snap Expandable Plug. Turn the Water Tee Valve in the direction of water flow, to the sample cup, to open. To replace the Quick Snap Expandable Plug, insert the plug back into the water tee hose, turn the lever clockwise to expand the plug against the sides of the tube, then snap the lever down to form a positive seal.

2–7–2 pH TESTER OPERATION (continued)



WATER TEE OPERATION
ILLUSTRATION 2–9

2–7–2 pH TESTER OPERATION (continued)



PH TESTER FRONT AND BACK VIEWS
ILLUSTRATION 2–10

4. Open, carefully, the Water Tee Valve and collect a minimum of a 1 1/2 inch (38mm) sample of coolant water into the Sample Container.

5. Close Water Tee Valve and replace the Quick Snap Expandable Plug.

Note
Do not immerse the PH meter above display level.

6. Immerse the pH Meter into the sample to a depth of approximately 1 1/2 inches (38mm).

7. Stir the sample, gently, and wait a few seconds for the reading to stabilize. Reading should be between 6–8.

8. Troubleshoot in conformance with vendor compressor manual if readings fall outside this range.

9. Turn off meter using ON/OFF switch.

10. Rinse the tip of the electrode in distilled or ionized water, blot dry, then replace protective cap.

2–7–3 DISSOLVED SOLIDS TESTER CALIBRATION

Description

The Dissolved Solids Tester (DiST) has a range of 10 to 1990 PPM with a resolution of 10 PPM. The DiST automatically adjusts readings by 2% per degree change in temperature from 20° C to give the correct reading. To change the batteries, carefully pull out the battery case and replace batteries (4 x 1.4volt Duracell MP675H or equivalent). See illustration 2–10 for location of Calibration Trim Pot and Battery Case.

Note

Containers for both the Calibration Solution and water sample are included in the Shield Cooler Test Kit.

1. Clean DiST Tester Electrode and Container (included in the Shield Cooler Test Kit) with distilled water, then dry.
2. Add enough DiST Meter Calibration Solution (GE P/N 2100661), into the clean container, to cover the DiST Tester Electrode.

Note

The Calibration Solution is rated in micro Siemens. To find the PPM value simply divide this number by 2 (e.g., if the Calibration is rated at 1413 micro Siemens, the PPM value would be 1413/2 = 706.5PPM).

3. Establish the PPM value of the Calibration Solution. This information is located on the label of the Calibration Solution.
4. Immerse the DiST Meter into the solution and adjust the Calibration Trimmer, located on the rear of the meter, to indicate the value determined in Step 2.
5. Turn the meter off using the ON/OFF switch.
6. Rinse the electrode in alcohol for a few minutes.
7. Clean the Container in distilled water, then dry.
8. Replace the protective cap on the DiST Meter.

2–7–4 DISSOLVED SOLIDS TESTER OPERATION

1. Remove the protective cap from the bottom of the DiST meter.

2. Turn on the meter using the ON/OFF switch.

3. Place a clean Sample Container at the end of the Water Tee Sample Line. See Illustration 2–9.

Note

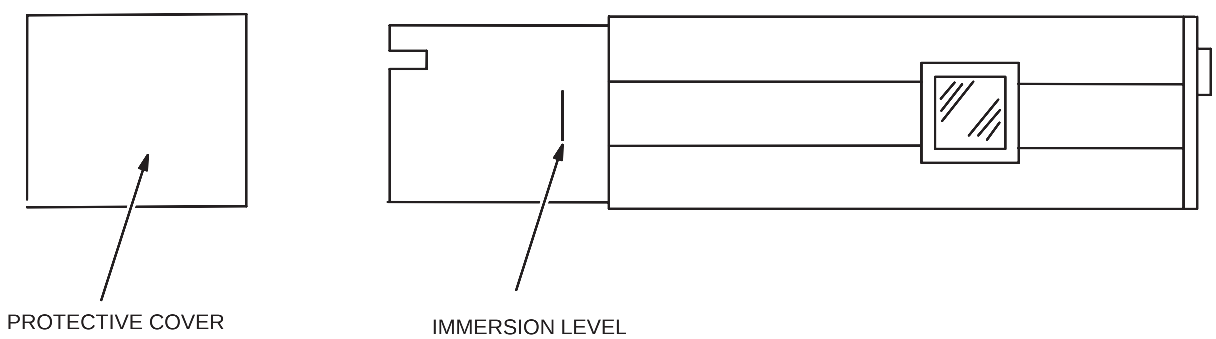
The Water Tee is designed with a safety Quick Snap Expandable Plug. To take a water sample, pull the lever up and turn counterclockwise and remove the Quick Snap Expandable Plug. Turn the Water Tee Valve in the direction of water flow, to the sample cup, to open. To replace the Quick Snap Expandable Plug, insert the plug back into the water tee hose, turn the lever clockwise to expand the plug against the sides of the tube, then snap the lever down to form a positive seal.

4. Open, carefully, the Water Tee Valve and collect a minimum of a 1 1/2 inch (38mm) sample of coolant water into the Sample Container.

5. Close Water Tee Valve and replace the Quick Snap Expandable Plug.

Note

Do not immerse the DiST meter above display level.



DIST METER
ILLUSTRATION 2–11

6. Immerse the DiST Meter into the sample to a depth of approximately 1 1/2 inches (38mm).

7. Stir the sample, gently, and wait a few seconds.

2–7–4 DISSOLVED SOLIDS TESTER OPERATION (continued)

Note

Readings will normally vary momentarily before stabilizing. Batteries could be in need of replacement if this does not occur or if the readings are unstable.

Note

When DiST is first introduced to a solution, the readings might decrease or increase for some time depending on whether the temperature is above or below 25° C.

Note

Specific amounts of Calcium Carbonate and Magnesium Carbonate (specified in vendor compressor manual) contamination are of concern. The dissolved solids tester gives a rough estimate of <u>all</u> salts in the cooling water. If the DiST reading is greater than 100 ppm, it is advised that a laboratory analysis of the sample be performed to determine if either the Calcium Carbonate or Magnesium Carbonate are out of the specified range.

8. Record meter reading after two minutes, to allow the temperature sensor to fully compensate for the temperature variance. Troubleshoot compressor cooling water system in conformance with vendor manual, if a reading greater than 100 ppm is observed.

9. Turn off the DiST Meter using the ON/OFF.

10. Rinse the electrode in alcohol for a few minutes.

11. Clean the Container in distilled water, then dry.

12. Replace the protective cap on the DiST Meter.

2–7–5 UV LIGHT TESTER OPERATION

Description

Ultraviolet energy cannot be detected by the human eye. A bluish light will be visible through the filter of the lamp. This is due to the emission of visible light from the tube. The special filter eliminates most of this visible light interference. The Shortwave lamp filter has a rated life of 1000 hours. When the ultraviolet intensity on the shortwave lamp decreases considerably, a new filter is needed. If the UV Light Tester is not operating properly, return to SPO for repair.

Operation

1. Turn off compressor and disconnect Helium Line to high pressure side of compressor.
2. Plug the UV Light into an AC outlet.

WARNING!

ALWAYS WEAR THE UV LIGHT PROTECTION GOGGLES TO PROTECT AGAINST DANGEROUS ULTRAVIOLET RAYS, WHEN OPERATING THE UV LIGHT.

3. Wear the UV Light Safety Goggles.
4. Press the red button on the handle of the UV Light.
5. Hold UV Light up to end of helium line disconnected in Step1.

Note

If oil contamination is observed in either Step 5 or Step 8, contact the National Support Center for isolation of all contaminated parts in the cryocooler system.

6. If a fluorescent blue appearance is observed, in the end of the helium line, there is oil contamination and one or more of the cryocooler components must be replaced.
7. Press the black button on the handle of the UV Light.

Note

The Aeroquip Connector used in Step 8 is found in the Shield Cooler Installation Kit P/N 46–281088G3.
The compressor may need to be recharged after performing Step 8, below.

8. Screw a male Aeroquip Connector into the helium line, removed in Step 1, for about 3 seconds then remove the aeropquip connector from the helium line. Allow the air, and any oil contamination to collect on A hard dry surface.
9. Press the red button on the handle of the UV Light.
10. Hold the UV Light to any substances released in Step 9 above.
11. If a fluorescent blue appearance, is observed, there is oil contamination and one or more of the cryocooler components must be replaced.
12. Replace any contaminated components as directed by the National Support Center.
13. Press the black button on the handle of the UV Light.
14. Make sure the compressor is operative after any necessary repairs are made and before leaving the site.

2–8 LIQUID HELIUM TOP FILL

DESCRIPTION

This procedure covers “Top Filling” of SII / SIII and MR MAX Magnets. Top fills may be performed on above magnets except when the magnet is empty where a “Bottom Fill” is required.

2–8–1 EQUIPMENT

TOP FILL EQUIPMENT

Liquid Helium Transfer Line 46–294512P1 3658 mm (12 feet) or 46–294512P2 2438 mm (8 feet)

17.00 inch (432 mm) Liquid Helium Top Fill Cryostat Stinger Assembly 46–294512P25

250 liter / 500 liter Dewar Stinger Assembly 46–294511P1 / P2

Top Fill Adapter Assembly 46–318197G1

Heat Gun TC402274

Regulator 46–258151P1

Cryogen Safety Kit 46–271137G1

Hose Assembly 46–271135P16 (20 ft.) or 46–271135P1 (40 ft.)

SEE ILLUSTRATION 2–12

WARNING!

MAKE SURE PROPER CRYOSTAT STINGER ASSEMBLY LENGTH IS USED (17.00”). EXTENDED LENGTH STINGER COULD CONTACT RAMP LEAD PINS AND TRANSMIT HEAT TO QUENCH THE MAGNET. SHORTER LENGTH STINGERS COULD RESULT IN DAMAGE TO BAFFLES IN VERTICAL STACK DURING FILL.

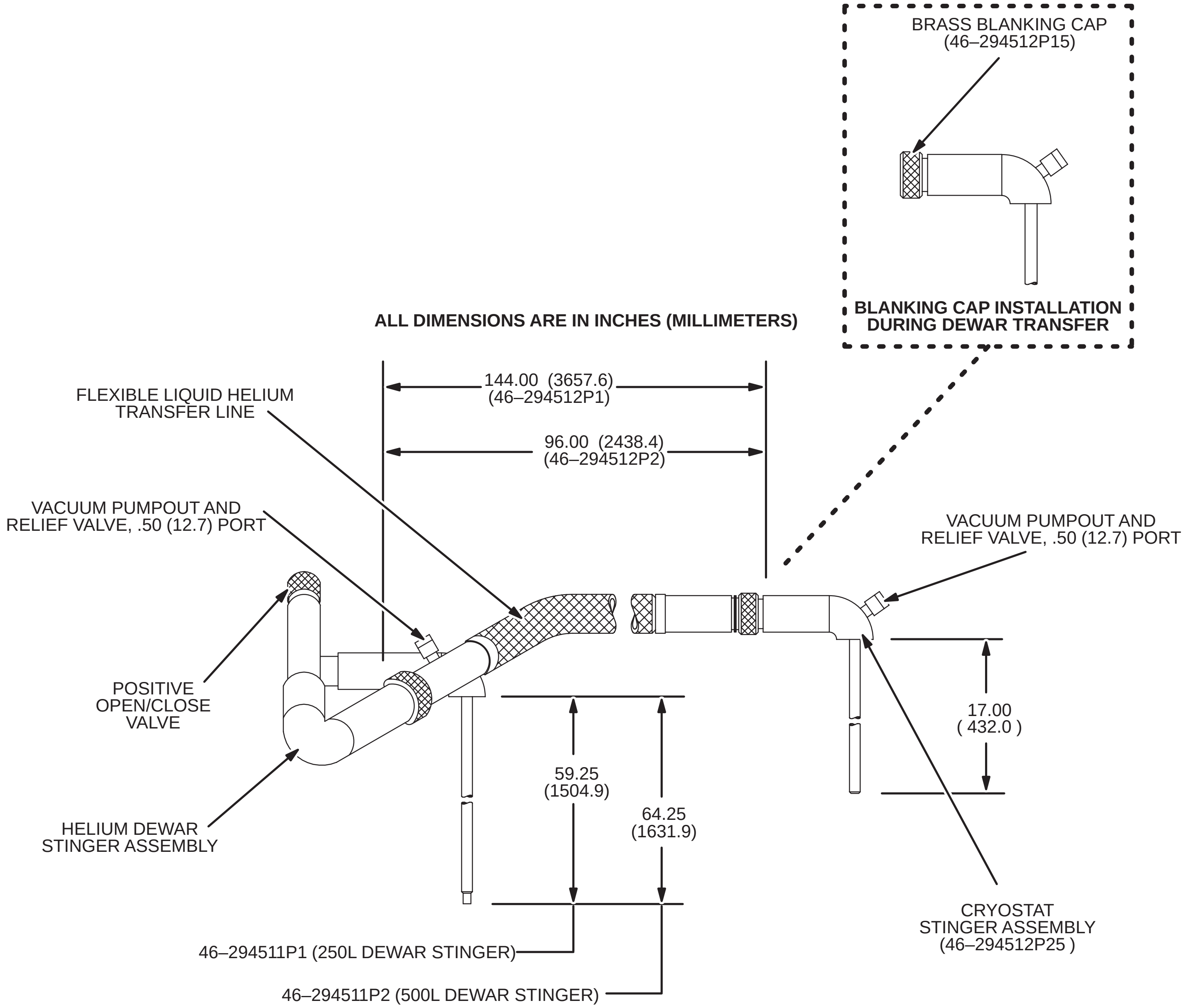
NEVER BRING HELIUM DEWARS, OR GAS CYLINDERS THAT ARE MADE FROM FERROMAGNETIC MATERIAL INTO THE EXAM ROOM. FERROMAGNETIC OBJECTS WILL BECOME DANGEROUS PROJECTILES IN A STRONG MAGNETIC FIELD. MAKE SURE ALL EQUIPMENT AND TOOLS USED IN THE EXAM ROOM ARE NON–FERROMAGNETIC.

SKIN CONTACT WITH LIQUID CRYOGENS WILL CAUSE BURNS. WEAR PROTECTIVE CLOTHING, GLOVES (NON–ABSORBENT MATERIAL) AND GOGGLES OR FACE SHIELD WHEN TRANSFERRING CRYOGENS.

MAKE SURE SUFFICIENT VENTILATION EXISTS IN THE EXAM ROOM TO DISPEL THE LARGE AMOUNTS OF HELIUM GAS WHICH WILL DISPLACE THE AIR (OXYGENS) AND COULD CAUSE ASPHYXIATION. VENT HELIUM FROM ROOM DURING LIQUID HELIUM FILL PROCEDURE.

SMOKING IS PROHIBITED IN THE EXAM AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN THUS PRODUCING A HIGHLY ENRICHED OXYGEN LIQUID.

2–8–1 EQUIPMENT (continued)



TOP FILL CONFIGURATION

VACUUM JACKETED HELIUM TRANSFER LINE AND DEWAR/CRYOSTAT STINGER ASSEMBLY

ILLUSTRATION 2–12

Procedure:

2–8–2 PREPARATION

CAUTION

Make sure the Helium Cryogen Level Meter is calibrated to enable accurate monitoring of LHe fill and the Shield Cooler is operating properly to minimize helium loss. If problems with Cryogen Meter or Shield Cooler exist, contact service.

1. Check cryogen meter calibration and record liquid helium level. Refer to vendor manual for cryogen meter calibration.
2. Obtain full liquid helium dewar. Check dewar pressure gauge. If pressure is above 1 psig, slowly open Dewar Vent Valve and reduce dewar pressure to 1 psig. See Illustration 2–13.

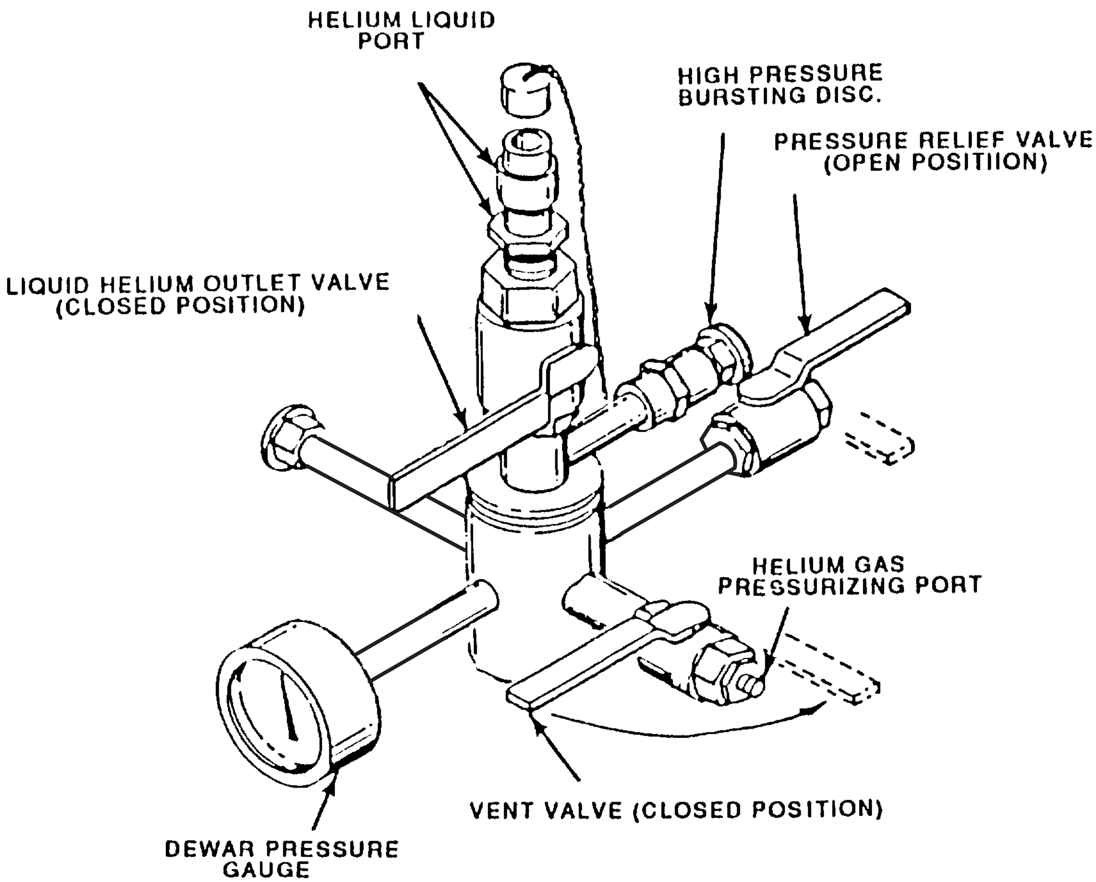
WARNING!

IF DEWAR PRESSURE DOES NOT VENT DOWN TO 1 PSIG, VERIFY THAT DEWAR PRESSURE RELIEF VALVE IS LEFT IN THE OPEN POSITION. CONTACT CRYOGEN SUPPLIER IMMEDIATELY.

Note

The Pressure Relief Valve is normally open during shipping and storage to prevent excessive build up of pressure in the dewar. Therefore, always leave Pressure Relief Valve open after using dewar.

2–8–2 PREPARATION (continued)



DEWAR CONNECTIONS
ILLUSTRATION 2–13

2–8–2 PREPARATION (continued)

3. Obtain 1 full GHe aluminum cylinder (135 SCF) for every 2 liquid helium dewars (250 liter) required.

WARNING!

SECURE CYLINDER BEFORE REMOVING PROTECTIVE VALVE CAP TO PREVENT CYLINDER FROM FALLING, WHICH COULD RESULT IN SHEARING VALVE OUTLET AND CAUSING HAZARDOUS HIGH PRESSURE GAS RELEASE.

4. Connect standard GHe Regulator And Hose Assembly to Valve Outlet (CGA 580) on GHe cylinder.

5. Make sure that Regulator Adjusting Knob is fully backed out, then slowly open GHe Cylinder Valve.

Note

If 99.999% Helium Gas is used, the purity of the gas remaining in the cylinder will degrade as a result of this process (i.e. the purity of the gas will be something less than 99.999%).

6. Observe Regulator High Pressure Gauge. Make sure indicated pressure is approximately 2000 psig indicating full cylinder.

7. Observe Cryostat Pressure Gauge and vent by temporarily opening Helium Vent Valve (V2) as required to obtain 0.3 – 0.5 psig pressure.

WARNING!

DO NOT PUT FACE DIRECTLY OVER THE RAMPING LEAD PORT. THE EXPULSION OF CRYOGENICALLY COLD HELIUM GAS CAN OCCUR UPON UNCAPPING OF THE RAMP LEAD PORT. ADDITIONALLY, A RAPID EXPULSION OF CRYOGENICALLY COLD HELIUM GAS WILL OCCUR IF THE MAGNET QUENCHES.

2–8–2 PREPARATION (continued)

8. Make sure Top Fill Adapter (46–318197G1) is installed in Ramp Lead Port. If not installed, install per the following steps (See Illustration 2–14).

   a. Vent magnet pressure below 0.25 psig.

WARNING!

MAKE SURE MAGNET PRESSURE IS BELOW 0.25 PSIG TO PREVENT CRYOGEN BURNS OR LOW OXYGEN LEVELS.

   b. Prepare the Top Fill Adapter with valve in the open position and the plug removed from the top of the adapter.

Note

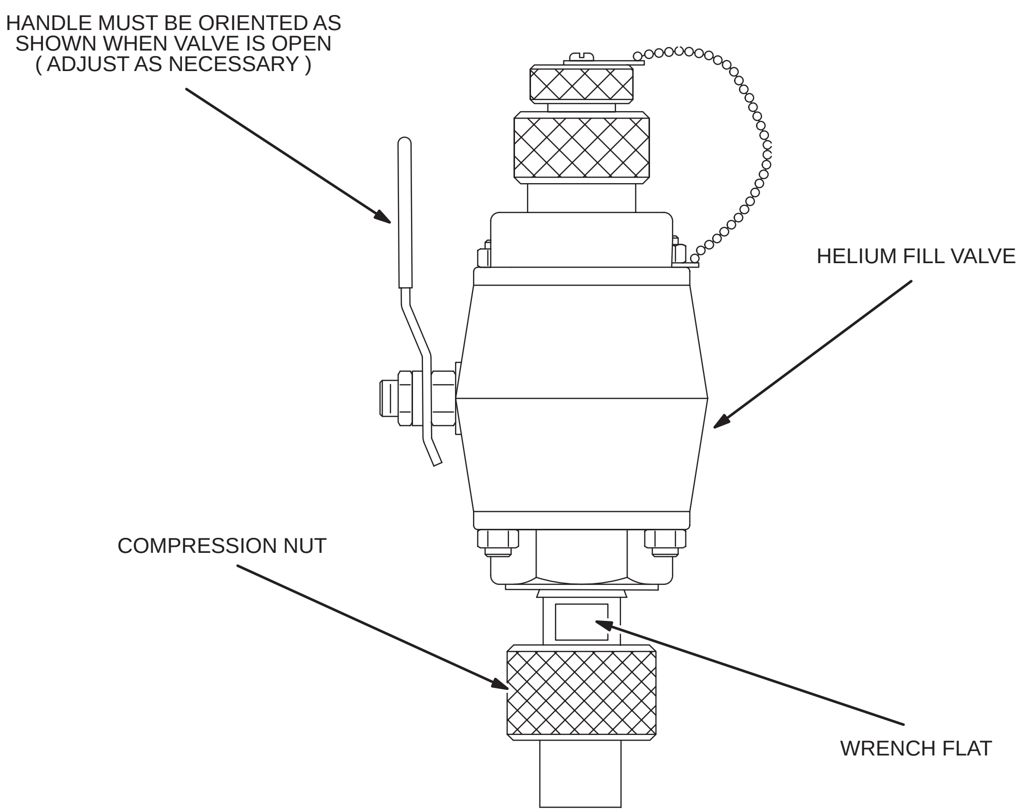
Perform Steps c and d quickly to prevent helium from filling the ceiling area and possible cryopumping.

   c. Remove one of the Ramp Port Caps and quickly insert the Top Fill Adapter securing it tightly to the magnet with the Compression Nut.

   d. Close the valve on the Top Fill Adapter.

   e. Insert the plug into the Top Fill Adapter.

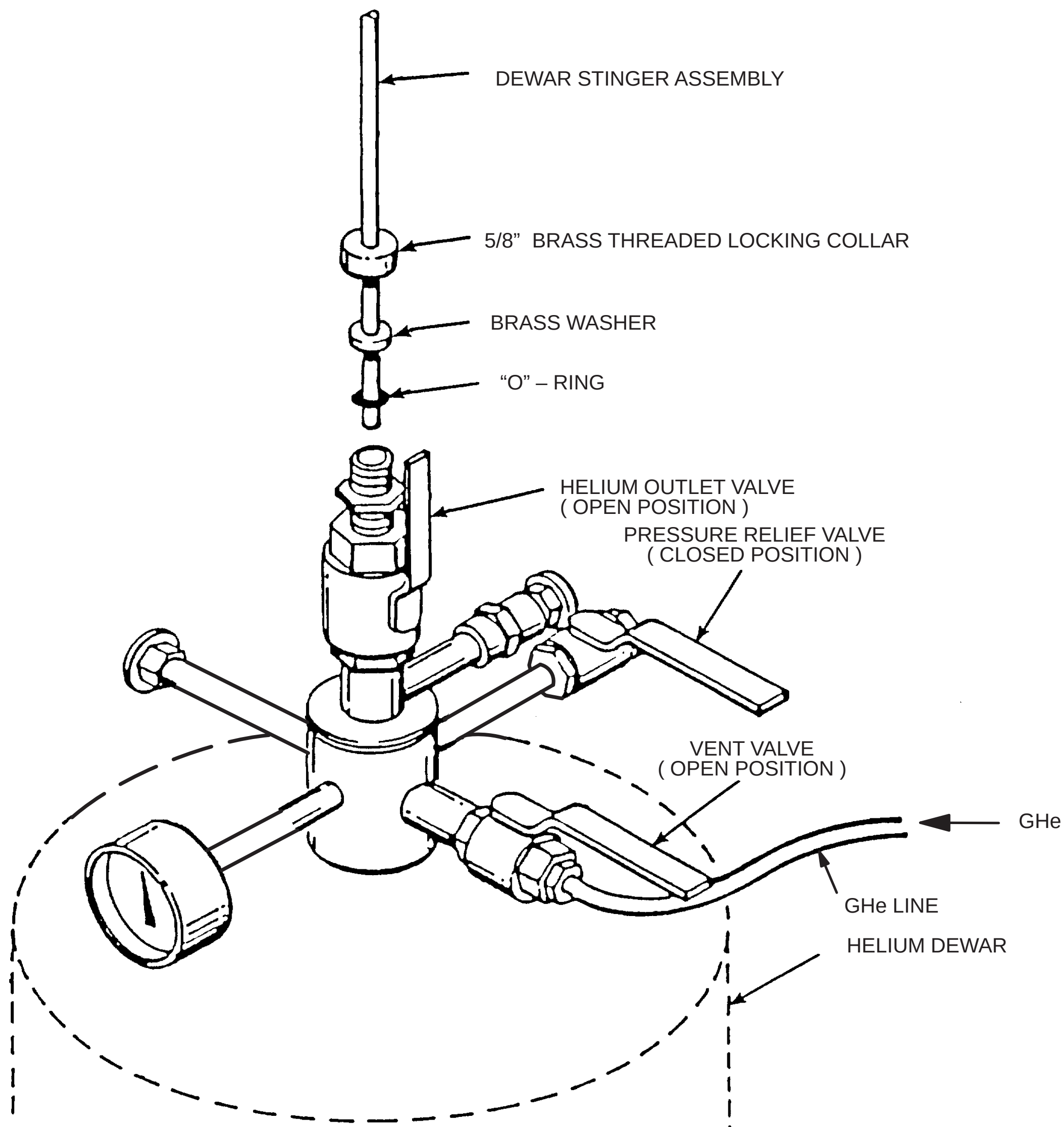
9. Make sure Top Fill Adapter is securely installed with no leaks.



TOP FILL ADAPTER
ILLUSTRATION 2–14

2–8–3 LIQUID HELIUM TOP FILL

1. Verify Dewar Helium Outlet Valve is in the closed position.
2. Loosen 5/8 inch Locking Collar.
3. Remove 1/2 inch Cap and Adapters exposing 5/8 inch Brass Locking Collar.
4. Verify that Pressure Relief Valve is in the open position. See Illustration 2–15.
5. Make sure Dewar Stinger Assembly Valve is closed. Insert Dewar Stinger Assembly thru 5/8 inch locking collar until Stinger Tip contacts Helium Outlet Valve.
6. Open Helium Outlet Valve.



HELIUM DEWAR TRANSFER LINE EXTENSION CONNECTION
ILLUSTRATION 2–15

2–8–3 LIQUID HELIUM TOP FILL (continued)

7. Slowly insert Dewar Stinger Assembly into dewar until Stinger Tip contacts liquid helium (indicated by pressure increase on Dewar Pressure Gauge and expulsion of gas from Pressure Relief Valve Port).

8. Continue to insert Dewar Stinger Assembly at a rate that maintains a maximum 5 psig reading on the Dewar Pressure Gauge.

9. When Dewar Stinger Assembly contacts bottom of dewar, raise Stinger Assembly 1 inch and securely tighten 5/8” Threaded Locking Collar.

Note

If ceiling height prohibits insertion of Dewar Stinger Assembly into dewar, dewar must be moved to an area with higher ceiling height and transported back into exam room.

10. When dewar pressure stabilizes at 5 psig, close Pressure Relief Valve.

11. Attach Cryostat Stinger Assembly onto Helium Transfer Line.

WARNING!

MAKE SURE PROPER CRYOSTAT STINGER ASSEMBLY LENGTH IS USED (17.00”). EXTENDED LENGTH STINGER COULD CONTACT RAMP LEAD PINS AND TRANSMIT HEAT TO QUENCH THE MAGNET. SHORTER LENGTH STINGERS COULD RESULT IN DAMAGE TO BAFFLES IN VERTICAL STACK DURING FILL.

12. Attach opposite end of Helium Transfer Line onto Dewar Stinger Assembly.

WARNING!

FIRMLY HOLD UNATTACHED END OF HOSE WHILE PURGING REGULATOR AND GAS LINE ASSEMBLY TO PREVENT WHIPPING MOTION.

13. Open Vent Valve V2. Vent cryostat down to 0.5 psig.

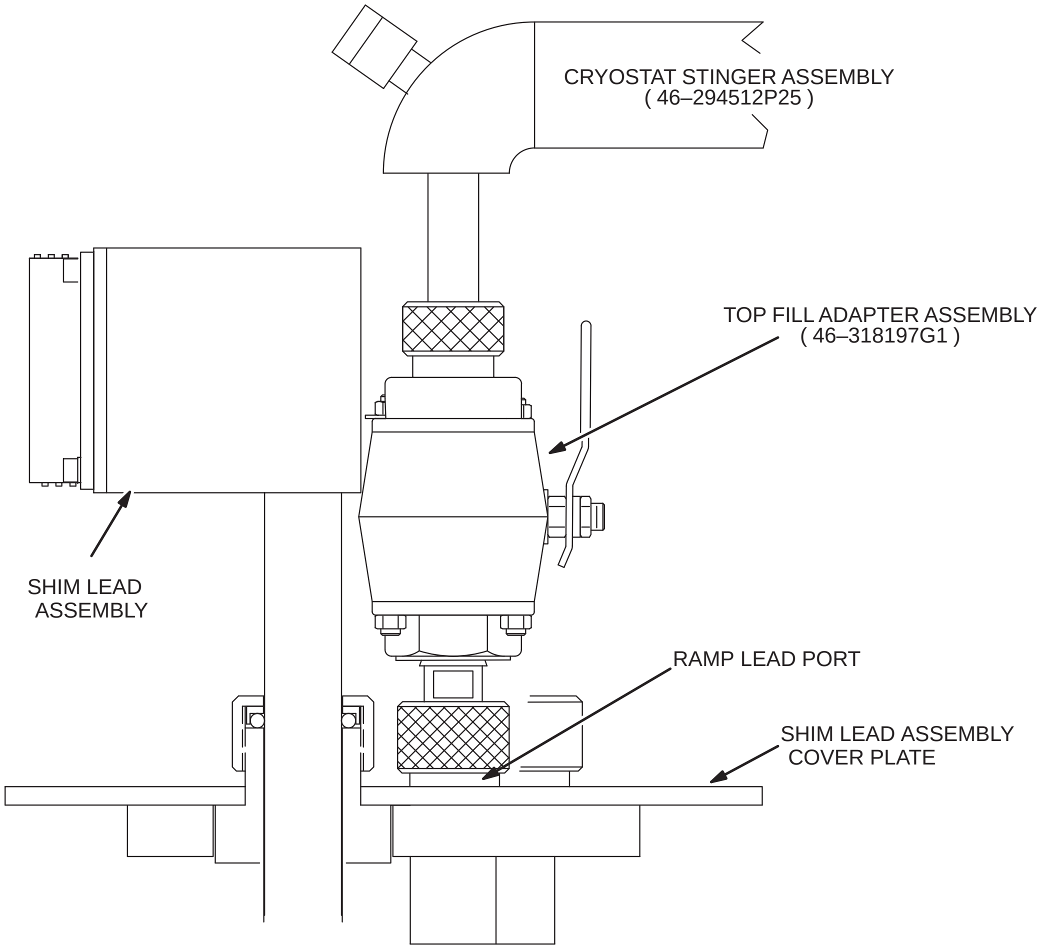
14. Purge GHe Regulator and Gas Line Assembly by alternately turning Regulator Handle fully in and out 3 times. Upon completion of purge, back Regulator out until minimal flow is felt exiting the Gas Line Assembly.

15. Open Dewar Vent Valve to allow a small amount gas flow while connecting gas line from GHe Cylinder. Attach purged Gas Line Assembly to full dewar Helium Gas Pressurizing Port while both the Dewar Vent Valve and GHe Cylinder have gas venting or flowing out. Fully back out Regulator Adjusting Knob.

2–8–3 LIQUID HELIUM TOP FILL (continued)

Note

Valve on Top Fill Adapter, installed on Ramp Lead Port, is refered to as "V1A" (V1 alternate) for purposes of this procedure. See Illustration 2–16.



TOP FILL ADAPTER ASSEMBLY INSTALLED IN MAGNET
ILLUSTRATION 2–16

CAUTION

Leave original Fillport Valve "V1" closed at all times during helium topfill to prevent cryopumping. Use "V1A" exclusively.

WARNING!

DO NOT INSERT A WARM CRYOSTAT STINGER FULLY INTO THE RAMP LEAD PORT. INSERTION OF WARM OBJECTS INTO THE CRYOSTAT CAN CAUSE A MAGNET QUENCH.

16. Loosen and remove Compression Fitting (Locking Collar, Washer and O–Ring) from V1A Fitting and insert onto Transfer Line Stinger. Save Plug.

2–8–3 LIQUID HELIUM TOP FILL (continued)

17. Partially open Dewar Stinger Assembly Valve allowing liquid helium to purge and precool Transfer Line Assembly until a liquid plume is observed exiting the Cryostat Stinger Assembly.

18. Slowly open alternate Fill Valve V1A

19. Insert the purged Cryostat Stinger into Fill Port approximately 8.00 inches (204 mm). Then assemble and tighten Compression Fitting slightly.

20. Slowly insert Cryostat Stinger fully into the Fill Port.

21. Tighten Compression Fitting.

Note

Check compression fitting tightness after 2 minutes. If gas is observed escaping from Compression Fitting on Top Fill Adapter or on Helium Dewar, use a heat gun to warm Compression Fitting and recheck compression fitting tightness.

22. Open Vent Valve at Helium Gas Pressurizing Port on dewar.

23. Verify GHe Cylinder Valve is fully open and adjust High Pressure Regulator Adjusting Handle to obtain a pressure which is 1.5 psig above the cryostat pressure during the entire fill.

CAUTION

Cryostat pressure will increase during fill. Make sure that 5 psig cryostat pressure is not exceeded during fill to prevent damage to Cryostat Pressure Gauge or Burst Disc.

Note

Helium Vent Port V2 should be frosting up and gas flow audible in the Helium Vent Line. If above conditions are not detected, a restriction/blockage in helium transfer circuit is indicated. If restriction /blocking is indicated, check for 1 inch stinger clearance from bottom of dewar or ice plugs in transfer line.

WARNING!

MONITOR DEWAR TRANSFER FOR AUDIBLE WHISTLING SOUND INDICATING THAT DEWAR IS EMPTY. DO NOT ALLOW AN EMPTY DEWAR TO BLOW WARM HELIUM GAS INTO RAMPED MAGNET AS A QUENCH COULD OCCUR.

24. Continue helium fill until 100% capacity (read on magnet cryogen level meters) is achieved. Monitor cryogen level readings during fill process every three minutes.

Note

Multiple dewars may be required to achieve 100% fill of magnet cryostat.

2–8–3 LIQUID HELIUM TOP FILL (continued)

25. Record information for each dewar on Magnet Fill Record (i.e. beginning / ending He level).
26. When cryostat is full (100%), or when changing helium dewars, close valve on dewar stinger assembly, close GHe Cylinder Valve, close Dewar Vent Valve, and open Dewar Pressure Relief Valve.
27. Disconnect Helium Transfer Line from Cryostat Stinger Assembly and immediately install Brass Blanking Cap onto Cryogen Stinger Assembly.
28. If additional dewars are required, change helium dewars in conformance with Section 2–8–4, "Changing Helium Dewars", prior to continuing with this procedure.
29. Monitor cryostat pressure. When cryostat pressure drops below 2.0 psig, close Helium Vent Valve V2 on magnet.

Note
A heat gun may be required to remove frost from the Top Fill Adapter assembly before removing Stinger.

30. Loosen Stinger Compression Fitting at V1A on Top Fill Adapter and quickly remove Cryostat Stinger Assembly. Immediately close V1A.
31. Replace Compression Fitting and Plug on Top Fill Adapter.
32. Disconnect Helium Transfer Line Assembly from Helium Dewar Stinger Assembly.
33. Remove Helium Dewar Stinger Assembly from helium dewar. Close liquid helium outlet valve.
34. Back off Pressure Regulator Adjusting Knob (CCW) on helium gas cylinder until no resistance is felt.
35. Disconnect Helium Gas Line from dewar Helium Gas Pressurizing Port.
36. Verify following dewar configuration.

A. Liquid Helium Outlet Valve	closed
B. Helium Vent Valve	closed
C. Pressure Relief Valve	open

D. Replace all adapters on Liquid Helium Valve Outlet

37. Remove GHe Cylinder Regulator from helium gas cylinder and install protective valve cap on cylinder.
38. Check and record cryostat pressure and Cryogen Level Meter readings. Ensure cryostat pressure is at 2.0 psig or below before leaving site.

2–8–4 CHANGING HELIUM DEWARS

1. Close Helium Vent Valve (V2) on magnet.

Note

If 99.999% Helium Gas is used, the purity of the gas remaining in the cylinder will degrade as a result of this process (i.e., the purity of the remaining gas will be something less than 99.999%).

2. Obtain full liquid helium dewar. Check dewar pressure gauge. If pressure is above 1 psig, slowly open Dewar Vent Valve and reduce dewar pressure to 1 psig. See Illustration 2–13.

WARNING!

IF DEWAR PRESSURE DOES NOT VENT DOWN TO 1 PSIG, VERIFY THAT DEWAR PRESSURE RELIEF VALVE IS LEFT IN THE OPEN POSITION. CONTACT CRYOGEN SUPPLIER IMMEDIATELY.

Note

The Pressure Relief Valve is normally open during shipping and storage to prevent excessive build up of pressure in the dewar. Therefore, always leave Pressure Relief Valve open after using dewar.

3. Observe GHe Cylinder regulator High Pressure Gauge. Make sure indicated pressure is at least 1000 psig indicating sufficient gas volume for transferring full 250 liter helium dewar.

WARNING!

SECURE CYLINDER BEFORE REMOVING PROTECTIVE VALVE CAP TO PREVENT CYLINDER FROM FALLING, WHICH COULD RESULT IN SHEARING VALVE OUTLET AND CAUSING HAZARDOUS HIGH PRESSURE GAS RELEASE.

4. If new GHe Aluminum Cylinder is required in Step 3, connect standard GHe regulator and hose assembly to valve outlet (CGA 580) on GHe cylinder.

5. Make sure that regulator adjusting handle is fully backed out, then slowly open GHe cylinder valve.

6. Refer to equipment in Helium Fill Section for appropriate equipment.

2–8–4 CHANGING HELIUM DEWARS (continued)

7. Observe cryostat pressure gauge and vent, temporarily opening Helium Vent Valve as required to obtain 1.0 psig pressure or below.
8. Verify Helium Outlet Valve is closed on full dewar.
9. Loosen 5/8 inch Locking Collar on full dewar.
10. Remove 1/2 inch cap and adapters exposing 5/8 inch Brass Locking Collar.
11. Verify that Pressure Relief Valve is in the open position.
12. Verify Dewar Stinger Assembly Valve is in the closed position. Disconnect Helium Transfer Line from Dewar Stinger Assembly.
13. Remove Dewar Stinger Assembly from empty dewar.
14. Wipe off frost or moisture on Dewar Stinger and insert Dewar Stinger Assembly thru 5/8 inch Locking Collar until stinger tip contacts Helium Outlet Valve in full dewar. See Illustration 2–16.
15. Open Helium Outlet Valve.
16. Slowly insert Dewar Stinger Assembly into dewar until Stinger Tip contacts liquid helium (indicated by pressure increase on dewar pressure gauge and expulsion of gas from pressure relief valve port).
17. Continue to insert Dewar Stinger Assembly at a rate that maintains a maximum 5 psig reading on the Dewar Pressure Gauge.
18. When Dewar Stinger Assembly contacts bottom of dewar, raise Stinger Assembly 1 inch and securely tighten 5/8 inch Threaded Locking Collar.

Note

If ceiling height prohibits insertion of Dewar Stinger Assembly into dewar, dewar must be moved to an area with higher ceiling height and transported back into exam room.

19. When dewar pressure stabilizes at 5 psig, close Pressure Relief Valve.
20. Attach Helium Transfer Line onto Dewar Stinger Assembly.

WARNING!

FIRMLY HOLD UNATTACHED END OF HOSE WHILE PURGING REGULATOR AND GAS LINE ASSEMBLY TO PREVENT WHIPPING MOTION.

2–8–4 CHANGING HELIUM DEWARS (continued)

21. Disconnect Helium Gas Line from empty dewar. Purge GHe Cylinder Regulator and Gas Line Assembly by alternately turning Regulator Handle fully in and out 3 times. Upon completion of purge, back Regulator out until minimal flow is felt exiting the Gas Line Assembly.

22. Slightly open He Gas Vent Valve to allow for He gas flow from dewar.

23. Attach purged gas line assembly to dewar Helium Gas Pressurizing Port. Fully back out Regulator Adjusting Handle.

24. Prepare empty dewar as follows:

    a. Liquid Helium Outlet Valve — closed

    b. Helium Vent Valve — closed

    c. Pressure Relief Valve — open

    e. Replace all adapters on Liquid Helium Valve Outlet

25. Proceed with “LIQUID HELIUM TOP FILL” , Section 2–8–3, Step 13.